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**T.E. (Computer Engineering) (Semester – I) Examination, 2014
DATA COMMUNICATION AND WIRELESS SENSOR NETWORKS
(2012 Course)**

Time : 3 Hours

Max. Marks : 70

Instructions : 1) Question **1, 2, 3, 4** (**10 marks each**). Solve either Question **1** or Question **2** and Question **3** or Question **4**.
2) Question **7** and **8** (**18 marks**). Solve **any one**.
3) Question **5, 6, 9, 10** (**16 marks each**). Solve either Question **5** or Question **6** and Question **9** or Question **10**.
4) **Neat** diagrams must be drawn **wherever** necessary.
5) Assume **suitable** data if **necessary**.
6) Figures to the **right** indicate **full** marks.

1. A) Encode the following binary data stream into unipolar, polar (RZ, NRZ), Bipolar NRZ and Manchester codes for given Data stream : 11100101. 5
B) A network of CSMA/CD has a bandwidth of 10 Mbps. If the maximum propagation time (including the delays in the devices and ignoring the time needed to send a jamming signal) is 25.6 μ s, what is the minimum size of the frame ? 5
2. A) What are different application of WSN ? 7
B) Give definitions. 1) Baud rate 2) Bit rate 3) SNR. 3
3. A) Explain what is meant by slope overload and granular noise distortion. Also explain how adaptive delta modulation improves system tolerance to slope overload. 7
B) Write short note on Virtual LAN. 3
4. A) Explain stop and wait ARQ, GO back-n ARQ and selective repeat ARQ. Comment on the performance of each. 6
B) Explain WSN architecture in details. 4
5. A) Write short notes on : 10
 1) S-MAC
 2) Sparse Topology and Energy Management (STEM).
B) Explain in detail contention based protocols. 6
6. A) With the help of detail flow schematic diagram explain slotted CSMA-CA protocol. 8
B) What do you mean by LEACH protocol ? Explain organization of LEACH protocol. 8

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| 7. A) What are different routing challenges and design issues in WSN ? | 8 |
| B) Explain in detail data dissemination and gathering. | 8 |
| C) What is piconet ? | 2 |
| 8. A) Write short notes on : | 10 |
| 1) Sensor Protocol for Information via Negotiations (SPIN). | |
| 2) Geographic and energy aware routing. | |
| B) What are different types of routing strategies in WSN ? | 8 |
| 9. A) Explain various parameters needed for determining distances to anchor nodes. | 8 |
| B) Explain the impact of anchor placement. | 8 |
| 10. A) Write short note on : | 8 |
| 1) Tiny OS | |
| 2) Magnet OS. | |
| B) Explain information based sensor tracking. | 8 |
